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26 September 1973

SUPERSEDING
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MILITARY SPECIFICATION

PROJECTILE, 20MM, HIGH PRESSURE TEST, M54A1

1. SCOPE

1.1 This specification covers the high pressure test projectile for use in Cartridge, 20mm, HPT, M54A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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2.1 (Cont)

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258859	- Projectile, 20mm, High Pressure Test, M54A1
IEL 7258859	- Index of Inspection Equipment Lists for: Projectile, 20mm, HPT, M54A1
F7258861	- Carton, for Inter-Plant Shipment of Projectiles, 20mm Assemblies and Metal Parts Assemblies (200)
D7259136	- Shipping Container for Inert Projectile, 20mm, Assemblies and Metal Parts Assemblies (10,000)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-64 -- Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General.-The projectile shall comply with the requirements of Drawing (Dwg.) D7258859, referenced specifications and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.2)

3.3 Banding.-The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat.

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3.4 Protective coating.

3.4.1 Phosphate coating.-The pretreatment process used shall comply with applicable requirements of Dwg. D7258859. The phosphate coating shall comply with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating.

3.4.2 Paint coating.-The paint coating shall comply with applicable requirements of Dwg. D7258859. Paint adhesion shall comply with applicable requirements of TT-C-490.

3.4.3 Corrosion resistance.-The painted surface of the projectile shall show no visible evidence of ferrous metal corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.5 Workmanship.-The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.5.1 Processing defects.-The projectile shall be free of dents, scratches, cuts, sharp edges, burrs and other forms of substandard workmanship.

3.5.2 Cleanliness.-The projectile and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample applicable to the approved process shall be submitted in accordance with contract requirements and shall consist of:

32 Rotating band blanks
125 Bodies
125 Projectiles, unpainted
125 Projectiles, painted
3 Standard test panels

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4.2.1 (Cont)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified by the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Quality Conformance Inspection.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification.-Each projectile and each packed projectile lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.3.2 Examination.-Examination for major and minor defects shall be performed as specified herein. Each component and each assembly found to be defective, shall be rejected.

4.3.2.1 Sampling and acceptance criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.3.2.1.1.c) In addition, where three or less defects appear in a classification, or, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used. (see 6.3)

4.3.2.1.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major ----------0.25%
Minor ----------0.40%

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4.3.2.1.1 (Cont)

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis.

Major	0.40%

4.3.3 Classification of defects 1/.-The classification of defects shall be as follows: (see 6.4)

4.3.3.1 Rotating band blank (see Dwg. B7258816, a detail of Dwg. D7258859).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Foreign matter	Visual
102. Defective metal (porosity, lamination, or inclusion)	Visual
103. Incomplete or deformed	Visual 2/
Minor:	
201. Width, max.	Gage

4.3.3.2 Body (see Dwg. D7258860, a detail of Dwg. D7258859).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Knurling missing, incomplete or deformed	Visual
102. Depth, knurling, min.	Visual/Gage 3/
103. Foreign matter in rotating band seat (at time of banding)	Visual (see 3.3)
Minor:	
201. Diameter, rotating band seat	Gage
202. Width, rotating band seat	Gage
203. Width, knurling, min.	Visual/Gage 3/
204. Length, nose to rear edge of rotating band seat	Gage

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4.3.3.3 Projectile, unpainted (see Dwg. D7258859 or D7258860, a detail of D7258859).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Diameter, bourrelet, min.	Gage
102. Diameter, body, rear of rotating band	Gage
103. Diameter, rotating band	Gage
104. Runout, rotating band with datum surface	Gage
105. Runout, body, rear of rotating band with datum surface	Gage
106. Location of crimp groove	Gage
107. Profile, crimp groove	Gage
108. Radius, rear edge of crimp groove, max.	Gage
109. Length, nose to rear edge of rotating band	Gage
110. Width, rotating band	Gage
111. Void, junction rotating band and body	Visual
112. Crack or split in body	Visual
Minor:	
201. Over-all length	Gage
202. Diameter, bourrelet, max.	Gage
203. Mutilated	Visual
204. Depth and width of undercut, front and rear of rotating band, max.	Gage
205. Finish	Visual
206. Base radius missing	Visual

4.3.3.4 Projectile, painted (see Dwg. D7258859).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Diameter, bourrelet, max.	Gage
102. Weight	Balance
103. Rotating band damaged	Visual
Minor:	
201. Paint not dry	Tactile
202. Marking incorrect, incomplete, illegible, or missing	Visual
203. Paint inadequate	Visual
204. Paint on rotating band	Visual
205. Foreign matter	Visual

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1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ A rotating band blank which is deformed or judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

3/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

4.3.4 Tests.-The tests listed in Table I shall be conducted on each lot in accordance with the test methods and procedures specified in 4.4. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

Table I

Test		Sample Size	Item	Rqmt Para
Band tightness	1/	315	Projectiles, painted	3.3
Phosphate coating (appearance and weight)	2/	3	Standard panels (see TT-C-490)	3.4.1
Paint adhesion	3/	13	Projectiles, painted (see TT-C-490)	3.4.2
Corrosion resistance	4/	32	Projectiles, painted	3.4.3

1/ If two or more units of the sample have an average radial void, between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void, greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

2/ Failure of the phosphate coating on the standard panels to comply with the applicable requirements, shall be cause for rejection of all projectiles phosphated since the preceding test.

3/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample, shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

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4.3.4 (Cont)

4/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample, shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.3.6 Inspection equipment.-Index of Inspection Equipment Lists (IEL) 7258859 identifies the applicable inspection equipment lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. B11075228, a detail of IEL 7258859).

4.4 Test methods and procedures.

4.4.1 Band tightness.-The method of test shall comply with all the requirements of the operating instructions for the band tightness tester (Dwg. B7259545, a detail of IEL 7258859). The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.4.2 Phosphate coating.-The method of test shall be as specified in TT-C-490 for the type of coating used (see Dwg. D7258859). The test panels shall be processed, with the projectiles to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.4.3 Paint adhesion.-The method of test shall be as specified in TT-C-490. The test shall be made on the painted surface of each unit of the sample. Visible exposure of underlying phosphate coating or ferrous metal of any unit of the sample as a result of the test, shall cause the unit to be classed defective.

4.4.4 Corrosion resistance.-The method of test shall be as specified in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectiles may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.4.3.

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5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Each lot of projectiles shall be packed as specified by Dwg. F7258861 or Dwg. D7259136.

5.2 Marking and labeling.-Marking or labeling of shipping containers shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders should specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection if not place of manufacture.

6.1.3 First article sample requirements (see 4.2).

6.2 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 Intermediate point inspection.-The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

NOTE: Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing Activity:

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